Automotive Cockpit and In-Vehicle Infotainment Development Starter Kit

Sr. No	Parameter	Required Specification
1	**Device Type**	High-performance automotive cockpit / in-vehicle infotainment (IVI) development kit – R-Car H3e-2G Starter Kit
2	**Processor Architecture**	64-bit ARM-based heterogeneous multicore processor
3	**Application CPU**	Quad-core ARM Cortex-A57, operating at 2.0 GHz or higher
4	**Additional CPU**	Quad-core ARM Cortex-A53, operating at approximately 1.2 GHz or higher
5	**Real-Time Processor**	Integrated dual-lockstep ARM Cortex-R7 real-time processor
6	**Cache Memory**	Cortex-A57: 48 KB L1 instruction + 32 KB L1 data + 2 MB L2 (shared across cluster); Cortex-A53: 32 KB L1 instruction + 32 KB L1 data + 512 KB L2 (shared across cluster); Cortex-R7: 32 KB L1 instruction + 32 KB L1 data
7	**GPU**	Integrated PowerVR-class 3D graphics accelerator (Series6XT-class) supporting OpenGL ES 3.1 or better
8	**Memory**	LPDDR4 SDRAM, minimum 4 GB, configured across 2 channels (64-bit wide combined), operating up to LPDDR4-3200 (1600 MHz)
9	**On-board Storage**	Minimum 128 GB eMMC or more
10	**Non-volatile Flash**	64 MB HyperFlash-class boot flash plus an additional 16 MB QSPI flash module
11	**Expansion Interface**	High-density 440-pin expansion connector providing SoC signals and high-speed/automotive peripheral interfaces
12	**Display Output**	Support for multiple simultaneous display outputs at SoC level (up to 4 channels), including HDMI and LVDS-class interfaces; HDMI available natively on-board
13	**Video Input**	Multiple high-speed video/camera input interfaces including MIPI-CSI2
14	**Video Processing**	Hardware-accelerated H.264/H.265-class video codec and image/video processing
15	**Ethernet**	Gigabit Ethernet with Automotive Ethernet AVB capability, RJ45 on-board connector
16	**CAN**	Minimum 2 × CAN-FD automotive communication interfaces
17	**PCIe**	Minimum 2 × PCI Express 2.0 interfaces (1-lane each)
18	**USB**	1 × USB 3.0 dual-role port plus up to 4 × USB 2.0 host/function/OTG interfaces

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19	**Audio**	Digital/analog audio interfaces including microphone and line/headphone audio support
20	**Storage Expansion**	MicroSD/SD host interface
21	**Debug Interface**	Dedicated JTAG debug interface (main SoC and programmable-logic JTAG)
22	**Serial Interfaces**	Multiple UART/serial communication interfaces plus dedicated on-board debug-serial connectors
23	**I²C / SPI**	Multiple I²C and SPI/synchronous-serial interfaces for peripheral expansion
24	**Hardware Security**	Dual-channel cryptographic engine supporting AES, DES, hashing and RSA acceleration
25	**Functional Safety**	Processor compliant with ISO 26262 (ASIL-B) functional-safety standard for automotive use
26	**Packaging**	Flip-chip BGA package, approximately 21 mm × 21 mm, 1384-ball, 0.5 mm pitch
27	**Board Form Factor**	Approx. 100 mm × 100 mm board; approx. 57 mm height including active cooling fan
28	**Operating Systems**	Support for Linux, Android and QNX-class automotive operating systems, with hypervisor/multi-OS (multivisor) support for concurrent guest operation
29	**Automotive Software**	Support for Automotive Grade Linux (AGL) / Yocto-based automotive Linux development
30	**Power Input**	External 5 V DC power input, rated approximately 4–8 A depending on configuration
31	**Application**	Automotive cockpit, IVI, digital instrument cluster, multi-display systems, multimedia and connected-vehicle prototyping
32	**Warranty**	Minimum One Year